Preventing SLF Contamination in Grape Harvests
Team: Grape Soup | Client(s): Cornell CALS Extension / E&J Gallo Winery / National Grape
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Context & Problem Statement
Our chosen problem is to decontaminate post-harvested vineyard batches by removing SLFs and debris from harvested grapes before they enter downstream juice or wine processing.

This problem selection was based on modularity, controllability, and impact. Rather than trying to address SLFs across the entire vineyard, we focused on a narrow, controllable stage after harvest where intervention is more feasible. Integrating a removal system into post-harvest processing is modular because it can fit into existing sorting or cleaning workflows, and high-impact because it affects the full harvested batch. Key constraints include compatibility with existing processing systems, maintaining grape quality, and minimizing additional labor or process complexity. Since grapes and SLFs differ in weight, density, size, and color, we can design mechanical systems to remove SLFs while preserving grape quality. If this problem is solved, wineries would have cleaner harvested fruit, less contamination in juice/wine processing, and reduced labor spent on manual sorting or cleanup.

Alternative sub-problem selections have been unsuccessful, such as attempts to prevent the SLFs from inhabiting the vines. Current harvesting and shaking methods are not selective enough to remove SLFs without also damaging fruit or reducing harvest efficiency, so insects can enter bins and downstream processing. By targeting the post-harvest stage, the system reduces SLF presence in the final harvested product regardless of how many insects were originally on the vine.

Final Prototype & Application:
We developed a density-based separation system installed at the vineyard's receiving area. Harvester bins' contents are emptied into a funnel that filters liquids from solids. The remaining solids (grapes and SLFs) then enter a water-filled flotation tank. Grapes weigh 3-5 grams while adult SLFs weigh fractions of a gram. Due to this difference, grapes sink while SLFs float and are skimmed from the surface with a rotary paddle mechanism. Clean grapes drain from the bottom to continue processing.

Our long-term vision is a standalone station integrated into winery receiving areas, sized to handle mechanical harvester outputs. The system would consist of an initial liquid separator, a flotation tank with continuous flow capability, an automated surface skimmer, and a bottom drain leading to existing processing equipment.

In the lab, we constructed a small-scale model to validate separation efficiency using water, grapes, and SLF simulants.

Figure 1: Density Surface Skimming Mechanism

Conclusion & Recommendation:
Overall, the prototype demonstrated that a buoyancy-based approach paired with a simple mechanical paddle wheel is promising and worth further development for separating SLF-like floaters, modeled with marker caps, from grapes quickly and repeatedly.

The system achieved strong removal performance across controllable operating speeds, and the observed tradeoffs (splashing at higher ROM and the possibility of grape agitation/carryover) point to targeted mechanical refinements rather than a substantial redesign of the separation method. This validates the core separation concept and provides a solid foundation for refining the mechanism into a more robust, demonstration-ready design.

Another key outcome is that plain water was sufficient to achieve the desired separation of the SLFs and grapes inside the box (sink/float behavior). This is beneficial because it avoids introducing sugar or other density modifiers, helping keep the grapes and process stream clean while reducing added contamination, residue, and cleanup burden.

The recommended next step is to refine the prototype for controlled disposal and continuous operation. Specifically, the next design should integrate a dedicated disposal valve at the back/bottom of the box to drain collected grapes, in a controlled manner, without stopping the process. Adding a valve enables a faster reset between trials, cleaner handling of grapes, and easier maintenance of consistent water levels during longer runs, especially at higher RPMs where displacement and spillage increase. After this refinement, the system should be integrated into the broader post-harvest workflow, particularly before juice extraction, and then scaled for higher-throughput testing.

Testing & Results:
Prototype testing validated the density-based separation mechanism and its performance relative to the defined success criteria. Grapes weighing approximately 3–5 g consistently sank, while SLF simulants, modeled with marker caps, remained buoyant and available for surface removal.

At low speed (~8.3 RPM), three trials yielded 4–6 out of 6 removals in 40–60 seconds, for an average removal efficiency of ~83%. At higher speed (~16.5 RPM), removal was 5 out of 6 in ~20 seconds, also ~83%. These results demonstrate that increasing rotational speed significantly improves processing time without reducing effectiveness. Across all trials, observed grape loss was 0%, indicating that the system removes SLF-like floaters without carrying grapes into the discard stream. This meets the target removal efficiency of 80% while exceeding the grape loss requirement of less than 2%.

The paddle wheel consistently engaged only the surface layer. In control tests with grapes only, no grapes were transferred to the discard compartment, confirming that the mechanism does not disturb submerged grapes and satisfies the surface interaction requirement.

The system operated through multiple consecutive trials without stalling or requiring manual clearing. A minor issue arose from the use of rigid marker caps, which occasionally became caught at the exit during

paddle rotation. This is likely a limitation of the simulants rather than the design, as real SLFs are deformable and less likely to interfere.

At higher speeds, increased water displacement and agitation were observed, creating potential risks for fluid instability and grape carryover in scaled systems. Overall, the system achieved reliable and effective separation across both operating conditions, with the main design tradeoffs being removal speed, fluid stability, and the future risk of unintended grape loss.

A brief qualitative observation suggested that short-term water exposure did not visibly affect grape appearance or firmness; however, further controlled testing is required.

EXPERIMENTAL RESULTS

Condition	Trial	Time (s)	SLFs Removed (out of 6)	Efficiency (%)
Low RPM	1	60	5 / 6	83%
Low RPM	2	50	6 / 6	100%
Low RPM	3	40	4 / 6	67%
High RPM	1	20	5 / 6	83%

Figure 2: Removal efficiency vs Speed Conditions

Prototype & Testing Details:
The design progressed through cardboard, wood, and 3D-printed prototypes, with each stage used to test whether the paddlewheel geometry and mounting system could reliably skim floating SLF simulants while leaving grapes behind. The cardboard prototype validated the basic box-and-paddlewheel concept, while the wood prototype enabled functional motion, interference, fastening, and cycling tests. In motion testing, the paddle rotated 1:1 with the crank for 10 full revolutions with no slipping, sticking, or binding. In interference testing, the paddle completed 10 full revolutions with no contact between the paddles, tank walls, chute, or bracket. In repeat-use testing, the crank and paddle were rotated at approximately 60 RPM for 1 minute, with no visible cracking, loosening, tape peeling, or increase in wobble.

These tests showed that the prototype met its core functional intent. However, the wood-and-tape mounting system showed partial translational movement. The final solution improves stiffness, alignment, and paddle repeatability while optimizing core skimming and separation capabilities.

Components List and Bill of Materials:

Item:	Component:	Description/ Function	Source	Qty	Unit Cost ($)	Total Cost ($)
1	Plastic Storage Container	Main flotation tank (4.5 gal)	McMaster (8993T46)	1	45.23	45.23
2	Container Lid	Tight-fitting lid (unused in final)	McMaster (8993T49)	1	9.19	9.19
3	3D Printed Bracket	Supports mounting to the container	3D Printed (PLA)	2	1.75	3.50
4	3D Printed Crank	Rotational input mechanism	3D Printed (PLA)	1	3.86	3.86
5	3D Printed Paddle	Surface skimming/ removal	3D Printed (PLA)	1	53.31	53.31
6	2-part Epoxy	Structural bonding (bracket - container)	Lab Stock	1	0.00	0.00
7	Super Glue	Bonding rubber to the paddle	Lab Stock	1	0.00	0.00
8	Rubber Scrap	Extends paddle length	Scrap Material	1	0.00	0.00

Total Cost = $115.09

Test Materials (Not Included in System Cost)

Component	Purpose	Source	Estimated Cost
Expo Marker Caps	SLF simulants (floaters)	Lab Stock	$0.00
Grapes	Represent harvested fruit	Team Provided	$0.00